Akul Sethi

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EDUCATION

Northeastern University 09/2022 — 04/2027
B.S. Mathematics and Physics, Minor in Computer Science
M.S. Applied Mathematics
GPA: 3.95/4.00
Honors: Summa Cum Laude, Best Research Capstone
Selected coursework: [akul-sethi.github.io/coursework.html]

PUBLICATIONS

Can It Edit? Evaluating the Ability of Large Language Models to Follow Code Editing Instructions
Conference on Language Modeling (COLM 2024)
Federico Cassano, Luisa Li, **Akul Sethi**, Noah Shinn, Abby Brennen-Jones, Jacob Ginesin, Edward Berman, George Chakhnashvili, Anton Lozhkov, Carolyn Jane Anderson, Arjun Guha
[paper: arXiv:2312.12450], [benchmark: huggingface.co/datasets/nuprl/CanItEdit]

PREPRINTS

Lower Bound for Region of Non-Distillable Magic States
Akul Sethi
[paper: arXiv:2610.37807]

The Random Purification Channel and Quantum Tomography
Akul Sethi, Scott McCary, and Theo Fairchild-Coppoletti
[paper: akul-sethi.github.io/reu.pdf]

EXPERIENCE

Northeastern University – Physics Department : Research Assistant 07/2026 — Present
Working on developing efficient quantum circuits for orthogonal polynomial transforms using the Hermite transoform. In particular, working on a O(log N) circuit for the Hanh polynomials. Supervisor: Ning Bao.

Northeastern University – Mathematics Department: Research Assistant 05/2026 — 07/2026
Worked on quantum tomography in research funded by the NSF. In particular, focused on understanding simpler and general constructions of the random purification channel. Supervisor: Eugene Tang.

Northeastern University – Mathematics Department: Teaching Assistant 09/2025 — 04/2026
Taught two semesters of mathematics for differential equations with linear algebra and introduction to mathematical reasoning. Held office hours, study sessions, and graded course work.

Schonfeld: Quantitative Researcher 01/2025 — 07/2025
Developed machine learning systems to model covariance of equity returns and detect anomalies in live trading systems. Read and implemented leading papers in the field to combine rigorous theory with empirical results.

KMW Technology: Software Engineer 01/2024 — 06/2024
Developed internal Retrieval-Augmented Generation tool combining traditional search with modern research developments in LLMs. Contributed over 250 commits to open-source search ETL using proper object-oriented design principles, version control, documentation, and unit-testing.

Northeastern University – Programming Research Lab: Research Assistant 09/2023 — 12/2024
Benchmarked large language model's ability for code editing in domain specific tasks. Developed RL based training methods and data preprocessing to improve models abilities in code editing. Received award from Northeastern Khoury College of Computer Science to support conference travel. Supervisor: Arjun Guha.

Last updated August 2026

PRESENTATIONS & POSTERS

- Can It Edit?: Evaluating the ability of Large Language Models to Follow Code Editing Instructions. Gave talk at International Conference on Software Engineering (ICSE) LLM4Code workshop [slides: akul-sethi.github.io/icse_talk.pdf].
- Lower Bound for Region of Non-Distillable Magic States. Gave talk to NEU mathematics department. [slides: akul-sethi.github.io/capstone_talk.pdf].
- The Random Purification Channel and Quantum Tomography. Gave talk to NEU mathematics department. [slides: akul-sethi.github.io/reu_talk.pdf].

AWARDS AND GRANTS

- Awarded Outstanding Math Research Capstone award. *05/26*
- Awarded $4000 for summer math research by NSF grant for Algebraic Geometry and Representation Theory. *05/26*
- Granted the Khoury College Travel Award ($1,500) to present at ICSE 2024. *03/24*

SKILLS

Programming Languages: Proficient: Python, Java, Racket Familiar: C++, C, SQL, HTML, CSS
Machine Learning & Scientific Computing: PyTorch, PyTorch Geometric, NumPy, pandas, scikit-learn, Matplotlib, Jupyter
Tools & Infra: Git, LaTeX, Docker, Poetry, Conda, Slurm, pytest, GitHub Actions, tmux